

Module 9 — Geant4 B4 Sampling Calorimeter

Supplementary to the sPHENIX Standalone Course

1. The B4 example family

B4 is Geant4's canonical sampling-calorimeter example: alternating absorber (Pb) and active (liquid argon) layers with per-layer energy readout. Four flavours ship in `examples/basic/B4`:

Variant	Scoring technique
B4a	User action classes (step-by-step bookkeeping)
B4b	Thread-local run action with accumulables
B4c	Sensitive detector with <code>G4VSensitiveDetector</code>
B4d	Scoring mesh via command interface

We use B4a because it's closest to what you'll write by hand later.

2. Geometry in one paragraph

World = 240 cm × 240 cm × 6 m air box. Inside: one logical 'calorimeter' volume, transverse 20 cm × 20 cm, 50 layers along z. Each layer = 2.3 mm Pb (absorber) + 5.7 mm liquid Ar (active). Layers are defined once and replicated with `G4PVReplica`.

3. Physics list

B4a uses FTFP_BERT by default. This is the recommended general-purpose hadronic list. For the course you don't need to change it; for later tuning:

List	Best for
FTFP_BERT	General HEP, ATLAS/CMS default
QGSP_BERT	High-energy LHC, older default
QGSP_BIC_HP	Medical physics, neutron precision
Shielding	Radiation studies
FTFP_BERT_EMZ	Higher-precision EM (slower)

4. Sampling fraction

In a sampling calorimeter, only the active layers are read out. The sampling fraction is:

$$f = \langle E_{\text{dep in active}} \rangle / \langle E_{\text{dep in active}} + \text{absorber} \rangle$$

For B4a's default geometry and 10 GeV electrons, $f \approx 0.14$. The measured response is therefore $E_{\text{measured}} = f \cdot E_{\text{incident}}$, and the 'calibrated' observable is E_{measured} / f .

5. Running a single energy

```
./exampleB4a -m run2.mac
# run2.mac:
/run/initialize
/gun/particle e-
/gun/energy 10 GeV
/run/beamOn 10000
```

6. Analysis output

B4a writes B4.root with a ntuple containing per-event Eabs (absorber energy), Eact (active energy), Labs, Lact (track length in each). Use TTree::Draw to plot:

```
tree->Draw("Eact/14.*1000:Eabs");  
// factor of 14%: your measured sampling fraction  
// scale Eact to incident-equivalent MeV
```

7. Longitudinal shower profile

For a richer study, add per-layer scoring and plot vs layer index. You should see a fast rise and a slow decay with a maximum around shower max:

$$\begin{aligned}t_{\text{max}} [X_0] &\approx \ln(E / E_c) - 1 \quad (\text{electrons}) \\t_{\text{max}} [X_0] &\approx \ln(E / E_c) + 0.5 \quad (\text{photons})\end{aligned}$$

With Pb ($X_0 = 0.56$ cm) and 2.3 mm per layer, each layer is $0.41 X_0$. So 10 GeV electrons ($E/E_c \approx 1300$) have $t_{\text{max}} \approx 6 X_0 \approx \text{layer } 15$.

8. Varying the physics list

To swap physics lists, edit B4a.cc:

```
G4VModularPhysicsList* physicsList = new FTFP_BERT;
physicsList->SetVerboseLevel(1);
runManager->SetUserInitialization(physicsList);
```

Swap FTFP_BERT for QGSP_BERT; recompile; compare resolution. You should see less than 2% difference for EM showers.

9. A worked sampling-fraction study

Goal: measure f and verify it is energy-independent.

Step	What to do
1	Run 10,000 e ⁺ at each of 1, 2, 5, 10, 20, 50 GeV
2	For each, compute $\langle E_{\text{act}} \rangle / \langle E_{\text{act}} + E_{\text{abs}} \rangle$
3	Plot f vs E ; should be flat within statistics
4	If f rises with $E \rightarrow$ leakage from the back of the calorimeter
5	Extend calorimeter depth until f is flat

10. Common issues

Symptom	Cause
Resolution tail at low energies	Physics-list secondary cuts too loose
Resolution tail at high energies	Transverse or longitudinal leakage
f decreases with E	Longitudinal leakage
f depends on primary particle	Hadronic showers sample differently from EM